HW-03 — Queue Models: Analysis and SimPy Implementation

Module M05 | Chapter 7 | Weight 10% | Difficulty ⋆⋆

Module: M05 Chapter: 7 Weight: 10% Due: End of Week 7

Overview

Analyze a 311 municipal call center analytically with the Erlang-C formula, verify the results in SimPy, extend the model with one advanced feature, and communicate your recommendation in a professional memo.

1 System

Parameter	Value
Arrival rate	$\lambda = 18$ calls/hr
Service rate	$\mu = 15$ calls/hr per operator
Service distribution	Exponential
Queue discipline	FCFS, infinite capacity

2 Parts

2.1 Part A — Analytical Analysis (20 pts)

For $c = 2$ and $c = 3$ operators, compute using the Erlang-C formula:

1. Traffic intensity ρ and offered load $a = \lambda/\mu$
2. $C(c, a)$ — probability that an arriving call must wait
3. W_q, W, L_q, L — all four performance measures

Present results in a single comparison table (rows: $c = 2$, $c = 3$; columns: ρ, C, W_q [min], W [min], L_q, L). Show the Erlang-C arithmetic for $c = 2$ in full; summarize for $c = 3$.

2.2 Part B — SimPy Implementation (40 pts)

Implement an M/M/c queue in SimPy with the following interface:

```python
def run_call_center(lam: float, mu: float, c: int,
                    sim_time: float = 10_000,
                    seed: int = 0) -> dict:
    """Returns dict with keys: Wq, W, Lq, L, utilization, n_served."""
    ...
```

Requirements:

- Run 30 independent replications for both $c = 2$ and $c = 3$
- Compute 95% t-based confidence intervals for W_q
- Verify: simulated W_q must be within 5% of the Erlang-C formula
- Plot simulated W_q (with CI error bars) alongside the analytical value

2.3 Part C — Extension (25 pts)

Choose **one** extension and implement it:

1. **Priority queues:** Add a priority class (e.g., emergency calls) using `simpy.PriorityResource`. Compare W_q for each class vs. the no-priority baseline.
2. **Reneging:** Callers abandon the queue after an exponential patience time (mean 3 min). Report the abandonment rate and its effect on W_q.
3. **Time-varying arrivals:** Use a two-period model: peak ($\lambda = 25$/hr, 8am–noon) and off-peak ($\lambda = 10$/hr, noon–5pm). Use NHPP thinning and plot W_q by hour.

2.4 Part D — Professional Memo (15 pts)

Write a 1-page memo (non-technical audience) addressed to the call center manager. It must state: the recommended staffing level, the simulated W_q with uncertainty, at least one limitation of the analysis, and one follow-up question the model cannot answer.

Grading Summary

Component	**Points**
Part A — Erlang-C analysis (both c values)	20
Part B — SimPy: runs, 30 reps, CI, verification	40
Part C — Extension (one of three)	25
Part D — Professional memo	15
Total	**100**

Full rubric: `rubrics/homework_rubric.pdf` | *Solutions: the instructor package (available to instructors on request).*